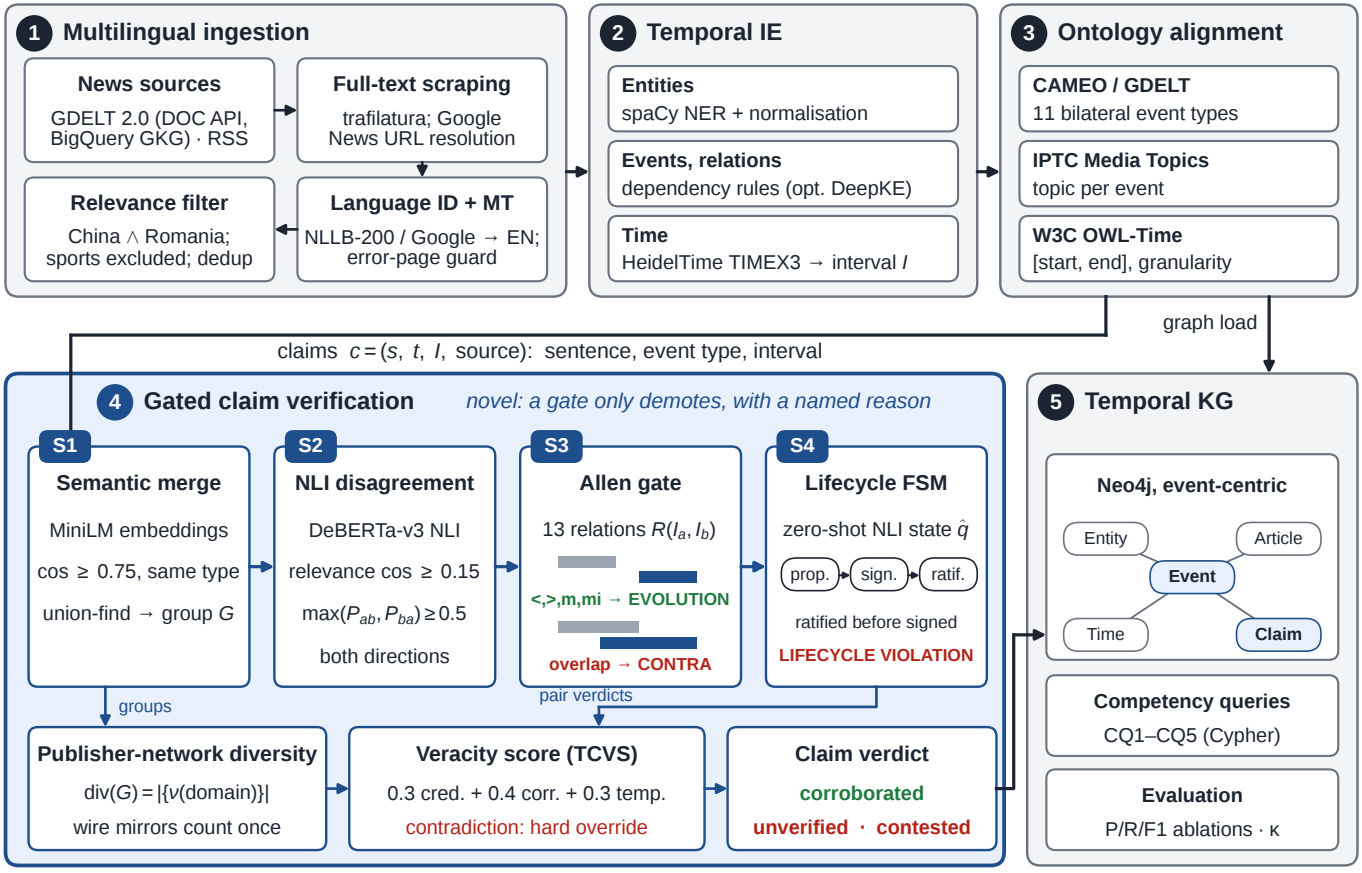

1 Multilingual ingestion
News sources
GDELT 2.0 (DOC API, BigQuery GKG) · RSS
Full-text scraping
trafilatura; Google News URL resolution
Language ID + MT
NLLB-200 / Google → EN; error-page guard
Relevance filter
China ∧ Romania; sports excluded; dedup
2 Temporal IE
Entities
spaCy NER + normalisation
Events, relations
dependency rules (opt. DeepKE)
Time
HeidelTime TIMEX3 → interval I
3 Ontology alignment
CAMEO / GDELT
11 bilateral event types
IPTC Media Topics
topic per event
W3C OWL-Time
[start, end], granularity
graph load
claims c = (s, t, I, source): sentence, event type, interval
4 Gated claim verification
novel: a gate only demotes, with a named reason
S1
Semantic merge
MiniLM embeddings
cos ≥ 0.75, same type
union-find → group G
S2
NLI disagreement
DeBERTa-v3 NLI
relevance cos ≥ 0.15
max(P_ab, P_ba) ≥ 0.5
both directions
S3
Allen gate
13 relations R(I_a, I_b)
<,>,m,mi → EVOLUTION
overlap → CONTRA
S4
Lifecycle FSM
zero-shot NLI state q̂
prop.
sign.
ratif.
ratified before signed
LIFECYCLE VIOLATION
groups
pair verdicts
Publisher-network diversity
div(G) = |{v(domain)}|
wire mirrors count once
Veracity score (TCVS)
0.3 cred. + 0.4 corr. + 0.3 temp.
contradiction: hard override
Claim verdict
corroborated
unverified · contested
5 Temporal KG
Neo4j, event-centric
Entity
Article
Event
Time
Claim
Competency queries
CQ1–CQ5 (Cypher)
Evaluation
P/R/F1 ablations · κ